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Key challenges in nurse-led foot care education for persons with diabetes: an international survey across six countries

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Abstract

Background Diabetic foot ulcers affect approximately one-third of people with diabetes during their lifetime and are a major cause of disability, amputation, and premature death. Preventing foot ulcers requires effective patient education. International guidelines have highlighted the central role of nurses in teaching routine foot care practices. However, nurse-led education varies considerably across healthcare systems, and it remains unclear which educational elements are consistently neglected internationally. This study aimed to identify common challenges in nurses' self-reported implementation of foot care education for persons with diabetes across six countries.

Methods A cross-national survey was conducted with 412 nurses in China, Japan, Kazakhstan, Lithuania, Nigeria, and Trinidad and Tobago, recruited via non-probability convenience sampling. Using the 15-item Diabetic Foot Care Education Scale nurse version, nurses rated how often they delivered each item on a five-point scale. Responses of 1–2 ("never/rarely") were classified as low implementation. The item-specific pooled benchmark for low implementation was calculated, and items exceeding the benchmark in ≥ 2 countries were identified as common challenges. Cross-country differences were examined using chi-square tests with Holm–Bonferroni adjustment.

Results Most participants were female (95.1%), with a median age of 43 years (IQR 35–51); 63.6% had more than 10 years of clinical nursing experience, and 73.8% had at least 3 years of diabetes care experience. Nine of the 15 items consistently identified low implementation across countries. In particular, footwear inspection and sock-fitting education differed significantly across countries (all $p_{\text{holm}} < 0.05$, Cramér's V ranged from 0.20 to 0.33). These findings indicate that key preventive elements of foot care, such as footwear inspection and appropriate sock selection, were not sufficiently addressed in nurse-led education for persons with diabetes.

Conclusions This multi-country study identified both substantial variability among countries as well as shared international gaps in nurses' implementation of foot care education. These findings indicate the importance of strengthening the implementation of commonly under-addressed educational elements while considering local

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contexts. By providing comparative evidence across diverse settings, this study may help inform context-sensitive educational strategies and support efforts to prevent diabetic foot ulcers.

Keywords Diabetic foot ulcers, Nurse-led education, Foot care education, Cross-national study, Preventive care

Background

Diabetic foot ulcers (DFUs) are a common and serious complication of diabetes mellitus (DM), affecting approximately 6.3% of people with diabetes, and are associated with a lifetime risk of 15–34% [1]. Diabetic foot ulcers are associated with increased risks of lower limb amputation and mortality, with 5-year mortality rates comparable to those reported for many cancers [2]. These ulcers significantly impair functional status and quality of life, leading to prolonged hospitalization, loss of productivity, and an estimated US\$10.9 billion in direct annual healthcare costs worldwide [3].

Preventing DFUs requires a multifaceted strategy. International guidelines emphasize regular foot screening and risk assessment, including assessment of diabetic peripheral neuropathy and peripheral artery disease, with preventive care tailored to risk level [4, 5]. In addition to clinical screening, structured foot care education is recognized as a core component of prevention, supporting daily self-care behaviors such as routine foot inspection, foot hygiene, safe nail and skin care, and appropriate footwear selection [5, 6].

Nurses are key providers of diabetes education in routine care and reinforce preventive messages [7]. In some settings, specialist diabetes or foot care nursing roles and structured education pathways are available, whereas in others, such roles and service organization are not universal [8, 9]. Even in well-resourced settings, sustaining ongoing foot care education for persons with diabetes in routine care remains challenging [10]. Together, these differences suggest that the delivery and sustainability of foot care education may vary across settings. However, it remains unclear whether similar challenges in the provision of nurse-led foot care education recur across health-care systems.

Moreover, the extent to which nurses address foot care education content differs and may be influenced by factors such as nursing training exposure [11] and attitudes toward DFU education [12]. Even when education is provided, it may lack sufficient specificity or fail to convey its importance effectively, limiting patients' understanding and acceptance of the information [13]. Taken together, these findings indicate that nurse-led foot care education varies not only in whether education is delivered, but also in which content is emphasized and how clearly it is communicated. However, prior studies remain fragmented across settings and rarely employ comparable item-level measurement of delivered content, making it

difficult to identify which specific education elements are consistently under-addressed across settings.

Importantly, recognizing cross-setting variation alone does not provide actionable targets for improvement. By revealing common blind spots across different contexts, this study provides empirical evidence to help nurses focus on the content areas most in need of reinforcement and support the sharing of best practices across health-care systems. Therefore, this study aimed to identify common challenges in nurses' self-reported implementation of foot care education for persons with diabetes, focusing on the specific foot care education elements that are most frequently under-addressed across countries.

Methods

Study design

This international cross-sectional study involved six countries: China, Japan, Kazakhstan, Lithuania, Nigeria, and Trinidad and Tobago. The selection of countries was primarily based on feasibility, including the availability of academic collaborators, partnerships, and the ability to obtain timely ethical approval.

Participants

Eligible participants were registered nurses working in clinical settings that provided care for persons with diabetes and who were directly involved in diabetes patient education. These settings included hospital endocrinology/diabetes departments, diabetes clinics, and internal medicine clinics. To improve cross-country comparability, the same eligibility criteria were applied across sites. Nurses working exclusively in managerial or administrative positions were excluded.

The sample size was determined using two complementary approaches. First, using G*Power for a chi-square test [14] (effect size $w = 0.2$, $\alpha = 0.05$, power = 0.8, $df = 5$), the minimum total sample size was 321. Second, to ensure the statistical validity and stability of country comparisons, this study applied a minimum of 30 participants per country, consistent with methodological recommendations [15]. Therefore, the recruitment target was ≥ 321 in total and ≥ 30 per country.

Recruitment

Nurses were recruited using non-probability convenience sampling in each participating country. Recruitment followed a common framework across countries, with site-specific adaptations to accommodate local health-care systems and professional networks. Participants

were primarily recruited through participating hospitals and healthcare facilities. In some countries, additional recruitment was conducted via professional nursing associations or networks, whereas in others, research team members directly approached eligible nurses at clinical sites. All participants were invited to complete the questionnaire after receiving study information and providing informed consent.

Data collection

Data were collected between February 2024 and July 2025 using standardized questionnaires. Both electronic and paper-based survey methods were employed, with procedures adopted to local contexts to maximize participation and maintain data quality. Electronic surveys were used in Japan and China, whereas paper-based surveys were predominantly administered in Kazakhstan, Trinidad and Tobago, and Nigeria. In Lithuania, a mixed approach was adopted, with paper questionnaires administered at the recruiting hospital and electronic surveys used for nurses from other institutions.

For electronic surveys, unique access links and reminder messages were used to enhance response rates and minimize duplicate submissions. For paper surveys, researchers provided standardized instructions, collected completed questionnaires, and reviewed them for completeness. Responses from paper questionnaires were entered into a database and independently verified by a second team member or the country principal investigator through comparison with the original forms prior to transfer to the analysis team.

Before participation, nurses were presented with a cover page describing the study purpose, anonymity of responses, data protection, and their right to decline participation without consequences. Informed consent was obtained prior to accessing the questionnaire, either by ticking a consent box on the paper questionnaire or by selecting an equivalent consent option in electronic surveys. Only participants who provided consent were able to proceed.

The questionnaire collected basic sociodemographic and work-related characteristics, including age, sex, total years of clinical nursing experience, and years of experience caring for persons with diabetes (DM care experience). It also included the Diabetic Foot Care Education Scale nurse version, which is described in detail in the Instruments section below.

Instrument Scale

The Diabetic Foot Care Education Scale nurse version (DFCES-N) was used to assess nurses' self-reported frequency of delivering foot care education to persons with diabetes. Nurses rated each item using a five-point Likert

scale (1 = never, 2 = rarely, 3 = sometimes, 4 = often, 5 = always), with higher scores indicating more frequent delivery of the corresponding educational content. The scale covers key foot care education elements, including foot inspection, skin and nail care, footwear selection, and related guidance [4].

The DFCES-N was adopted from the Turkish Foot Self-Care Behavior Scale (FSCBS), originally developed by Emine Kır Biçer and Nuray Enç to assess foot self-care behaviors among persons with diabetes [16]. The Turkish FSCBS consists of 15 items and has demonstrated acceptable internal consistency (Cronbach's $\alpha = 0.83$). The FSCBS items were originally derived from the Foot Self-Care Observation Guide developed in an intervention study [17]. With permission from the FSCBS authors and the developers of the source item set, all 15 items were reworded to reflect nurse-led educational behaviors rather than patient self-care behaviors. No new domains were added. The original FSCBS items and the corresponding nurse version are presented in Supplementary Table A1.

Expert panel review

A six-country multidisciplinary expert panel of 15 members, including diabetes nurse specialists, endocrinologists, and methodologists, reviewed the original FSCBS items to create a nurse version. The panel retained each item's preventive construct and scope, and reworded only the item wording to shift the perspective from patient self-care to nurses' delivery of the corresponding education, without adding, deleting, or re-ordering any domains (Supplementary Table A1). Conceptual equivalence to the original items was confirmed by panel consensus. Content validity of the nurse version was independently assessed by three trained diabetes-care nurses. They rated each item for relevance to nurse-led foot care education and for clarity and comprehensiveness [18]. The item-level content validity index was 1.0 for all items on both dimensions, and the scale-level index based on the average across items was 1.0.

Translation and cross-cultural adaptation of the scale

Translation and cross-cultural adaptation were conducted based on established recommendations [19, 20]. For countries where the native language is not English (China, Japan, Kazakhstan, and Lithuania), two independent forward translations were produced and reconciled by the country team into a single version. This version was then independently back-translated into English. The back-translated English version was compared with the original items to confirm semantic and conceptual equivalence. Discrepancies were documented and resolved with the country teams. Cognitive debriefing with local nurses was conducted in China, Japan, and Kazakhstan.

In Lithuania, issues identified during the equivalence review were resolved through team discussion. No conceptual discrepancies were identified. Only minor wording edits were made for clarity, including three items in the Chinese version. Country-specific procedures are provided in Supplementary Table A2.

In Nigeria and Trinidad and Tobago, English is the working language; therefore, the English nurse version was used. Nevertheless, each country's principal investigator and local researchers reviewed the wording to confirm clarity and contextual appropriateness. Minor wording edits were made after local review in Trinidad and Tobago, and no changes were made in Nigeria. No conceptual changes were made (Supplementary Table A2).

Internal consistency of the DFCES-N in the pooled six-country sample was assessed using Cronbach's α ($\alpha = 0.962$).

Data analysis

Data were analyzed using IBM SPSS Statistics version 30.0 (IBM Corporation) and R (R Foundation for Statistical Computing, Vienna, Austria). Descriptive statistics summarized participants' demographic and work-related characteristics. Group comparisons used the chi-square test for categorical variables and the Kruskal–Wallis test for continuous variables that were not normally distributed.

For descriptive purposes, country-specific mean scores and standard deviations (SDs) were calculated for each DFCES-N item. Full country-specific item response distributions (n, %) and mean (SD) values are presented in Supplementary Tables A3 and A4.

To identify educational elements that were not routinely delivered, low implementation was defined as responses of 1 ("never") or 2 ("rarely"), whereas responses of 3–5 ("sometimes", "often", "always") were classified as non-low implementation. Using pooled data from all six countries, an item-specific benchmark was defined as the overall proportion of low implementation in the total sample. Items for which the proportion of low implementation exceeded this benchmark in two or more countries were identified as potential common challenges. Item-level chi-square tests were performed to compare the distribution of low versus non-low responses across the six countries. To control for multiple testing across the 15 items, the family-wise error rate (FWER) was controlled at 0.05 using the Holm–Bonferroni method. Effect sizes were reported as Cramér's *V*. Country-specific rates of low implementation were visualized using a heat map.

Ethical considerations

This international study was conducted in accordance with the Declaration of Helsinki and approved by the

Ethics Committee of Nagoya University, Japan (Approval No. 2023–0439). Ethical approval was obtained from the relevant ethics committees of all participating countries before data collection. This study was reviewed and approved by the Ethics Committee of Lithuanian University of Health Sciences (Nr. 2024-BEC2-837), by the Ethics Committee of Guangdong Provincial Hospital of Chinese Medicine in China (BE2024-146) and under Protocol No. 26 dated March 28, 2025, in Kazakh National Medical University named after S. D. Asfendiyarov. In Nigeria, approval has been granted by multiple healthcare institutions, including the Nnamdi Azikiwe University Teaching Hospital (NAUTH/CS/66/VOL.17/VER3/40/2024/90) and the Chukwuemeka Odumegwu Ojukwu University Teaching Hospital (COOUTH/CMAC/SWS/Vol.11/001). In Trinidad and Tobago, ethical approval was obtained from the University of the West Indies (CREC-SA.2791/07/2024) and North, Central, and Eastern Regional Health Authorities of the country in 2024. All participants were provided with detailed information about the study aims, procedures, and data confidentiality. Participation was voluntary and anonymous. All data were handled confidentially and used exclusively for research purposes. The data collected in each country were stored securely and accessible only to authorized research team members. All data storage and handling procedures complied with the relevant data protection regulations and ethical standards of each participating country. Clinical trial number: not applicable.

Results

Basic characteristics of the six countries' nurses

A total of 412 nurses were recruited from China ($n = 73$), Japan ($n = 70$), Kazakhstan ($n = 70$), Lithuania ($n = 45$), Nigeria ($n = 97$), and Trinidad and Tobago ($n = 57$) (Table 1). Overall, participants were predominantly female (95.1%), with a median age of 43 years (IQR 35–51); 63.6% had more than 10 years of clinical nursing experience, and 73.8% had at least 3 years of diabetes care experience. The distribution of sex did not differ significantly across countries. Median age varied significantly among countries ($p < 0.001$), ranging from 35 years (IQR 27–38) in China to 53 years (IQR 45–57) in Lithuania. The three-level distribution of total clinical experience (< 7 years, 7–10 years, > 10 years) was similar across countries. In contrast, DM care experience showed significant cross-national differences ($p < 0.001$); the proportion of nurses with ≥ 3 years of DM care experience ranged from 51.4% in Kazakhstan to 95.6% in Lithuania.

Implementation of foot care education by country

In the total sample ($n = 412$), mean scores across the 15 DFCES-N items (nurse-reported frequency of addressing each item when providing education to persons with

Table 1 Basic characteristics of six countries' nurses ($n=412$)

		China ($n=73$)	Japan ($n=70$)	Kazakh- stan ($n=70$)	Lithuania ($n=45$)	Nigeria ($n=97$)	Trinidad and Tobago ($n=57$)	TOTAL ($n=412$)	χ^2	p	Cra- mér's Va/ ϵ^2b
		$n, \%$	$n, \%$	$n, \%$	$n, \%$	$n, \%$	$n, \%$	$n, \%$			
Age ($n=411$)	Median (IQR)	35 (27.0–37.5)	45 (27.0–51.3)	41 (30.3–50.0)	53 (44.5–57)	45 (40.0–54.0)	45 (40.3–50.0)	43 (35.0–51.0)	99.891	<0.001	0.23 ^b
Sex ($n=411$)	Male	8 (11.0)	5 (7.1)	2 (2.9)	0	4 (4.1)	1 (1.8)	20 (4.9)	10.813	0.055	0.16 ^a
	Female	65 (89.0)	65 (92.9)	68 (97.1)	45 (100.0)	93 (95.9)	55 (98.2)	391 (95.1)			
Total clinical experience	< 7 years	23 (31.5)	18 (25.7)	24 (34.3)	8 (17.8)	24 (24.7)	8 (14.0)	105 (25.5)	12.404	0.259	0.12 ^a
	7–10 years	10 (13.7)	5 (7.1)	5 (7.1)	6 (13.3)	11 (11.3)	8 (14.0)	45 (10.9)			
	> 10 years	40 (54.8)	47 (67.1)	41 (58.6)	31 (68.9)	62 (63.9)	41 (71.9)	262 (63.6)			
DM care years	< 3 years	18 (24.7)	18 (25.7)	34 (48.6)	2 (4.4)	30 (30.9)	6 (10.5)	108 (26.2)	37.583	<0.001	0.30 ^a
	≥ 3 years	55 (75.3)	52 (74.3)	36 (51.4)	43 (95.6)	67 (69.1)	51 (89.5)	304 (73.8)			

Note:

• Abbreviations: IQR, interquartile range; DM, diabetes mellitus

• Statistical tests: Age was compared with the Kruskal–Wallis test (effect size ϵ^2); Sex with Fisher's exact test; Total clinical experience and DM care experience with χ^2 tests (effect size Cramér's V)• a Cramér's V : range 0–1; ≈ 0.10 small, 0.30 medium, ≥ 0.50 large• b ϵ^2 (epsilon-squared): range 0–1; ≈ 0.01 small, 0.06 medium, ≥ 0.14 large

diabetes; Likert 1–5) ranged from 3.33 to 4.23. The highest mean score was observed for not using sharp instruments on feet (DFCES-N15; mean = 4.23), whereas the lowest mean score was observed for not applying lotion between the toes (DFCES-N04; mean = 3.33). Scores on the 15 DFCES-N items differed across countries (Fig. 1). China and Lithuania showed high mean scores with small standard deviations (SDs). In China, most education items ranged from 4.03 to 4.63, except for not applying lotion between the toes (DFCES-N04), which was lowest at 3.70; SDs were 0.7–1.4. In Lithuania, all education items ranged from 4.36 to 4.78, with SDs of 0.7–1.1. Japan and Kazakhstan showed moderate mean levels with greater variability. In Japan, education items ranged from 3.00 to 4.41, with SDs of 1.0–1.6. In Kazakhstan, item means were 3.39–4.14, with SDs of 1.3–1.5. Nigeria and Trinidad and Tobago showed lower overall scores and larger dispersions. In Nigeria, the mean scores varied widely from 2.69 to 4.37, and SDs were 1.0–1.5. In Trinidad and Tobago, mean scores were 3.35–4.05, and SDs were 1.5–1.6. Across all six countries, some education items showed relatively high mean levels, for example, not using sharp instruments on feet (DFCES-N15) and checking the soles for callus, rash, blister, wound (DFCES-N08). In contrast, not applying lotion between the toes (DFCES-N04) showed the lowest scores in multiple countries and the greatest between-country spread (2.69–4.36).

Common challenges in nurses' implementation across countries

Figure 2 shows the country-specific proportions of nurses with low implementation for each DFCES-N item,

together with Cramér's V and Holm-adjusted p values (p_{holm}). In the total sample ($n=412$), low implementation varied across items, ranging from 11.2% to 32.0%. The highest rate was observed for not applying lotion between the toes (DFCES-N04, 32.0%), whereas lower rates were observed for checking soles for callus, rash, blister, wound (DFCES-N08, 11.2%), and not using sharp instruments on feet (DFCES-N15, 11.4%).

Items whose country-specific low implementation proportion (scores 1–2) was higher than the pooled six-country proportion for the same item in ≥ 2 countries were considered potential common challenges. Between-country heterogeneity was evident for 14 of 15 items ($p < 0.05$; Avoid barefoot walking, DFCES-N10, $p = 0.052$). After Holm–Bonferroni adjustment, nine items remained statistically significant ($p_{holm} \leq 0.046$). These items mainly involved bath-water temperature, interdigital lotion use, toenail care, inspection of interdigital skin, footwear check, footwear characteristics, and sock fit (Fig. 2). For items with p_{holm} values < 0.05 , Cramér's V ranged from 0.20 to 0.33.

Four countries—Japan, Kazakhstan, Nigeria, and Trinidad and Tobago—had benchmark-exceeding low implementation rates for checking inside shoes for foreign matters like nail, dust or stone (DFCES-N09), and for ensuring that socks fit well (DFCES-N14). Three countries—Kazakhstan, Nigeria, and Trinidad and Tobago—shared low implementation of cutting toenails straight (DFCES-N05), checking toenails in terms of thickening, ingrowth and length (DFCES-N06), and wearing properly sized shoes (DFCES-N11), while Japan, Nigeria, and Trinidad and Tobago showed low implementation of not applying lotion between the toes (DFCES-N04), and

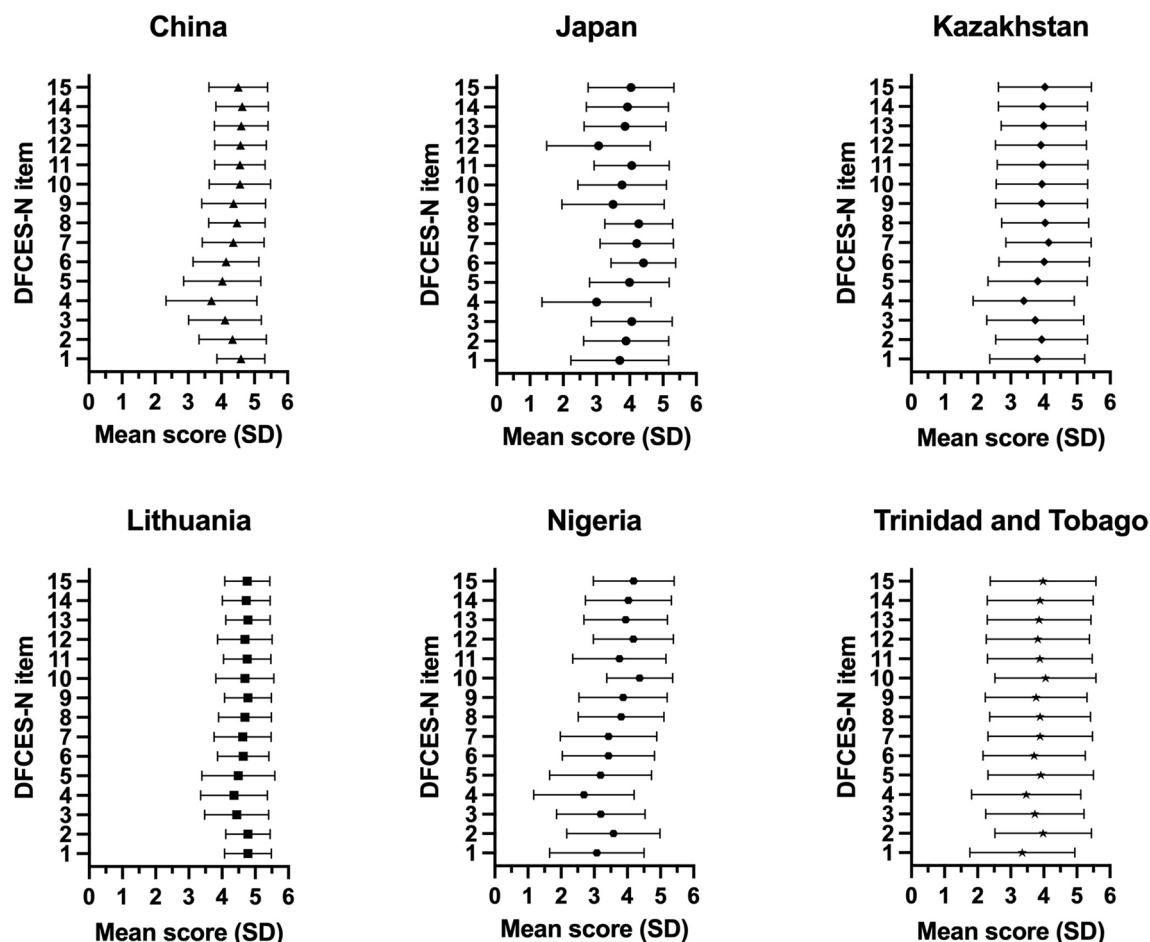


Fig. 1 Nurses' self-reported implementation of foot care education elements by country (Mean, SD, $n=412$)

Japan, Kazakhstan, and Trinidad and Tobago showed similar challenges regarding wearing shoes with soft leather and smooth internal surface (DFCES-N12). Additionally, two countries, Nigeria and Trinidad and Tobago, exhibited low implementation of checking the temperature of bath water (DFCES-N01) and of checking between toes for denudation, fungus and rash due to humidity (DFCES-N07).

Discussion

This study aimed to identify common challenges in nurses' self-reported implementation of delivering foot care education for persons with diabetes across six countries. The main finding was that the observed gaps were content-specific, with recurrent blind spots on toenail care, interdigital care, and footwear inspection across multiple countries, indicating that some elements are less consistently emphasized in routine nurse-led education. In contrast, core safety and sole inspection elements such as avoiding sharp instruments on the feet and checking the soles for lesions were more consistently reported, suggesting targeted blind spots rather than uniformly

low delivery of foot care education. Notably, the sock fit item displayed between-country heterogeneity despite relatively high mean scores, underscoring that frequent delivery on average may still mask uneven emphasis across settings. These findings provide international, item-level evidence that some elements of nurse-led foot care education are addressed, whereas several remain common challenges that single-country studies may not fully capture.

Footwear and socks

In this multi-country study, footwear inspection (DFCES-N09) and ensuring proper sock fit (DFCES-N14) emerged as the most consistently under-addressed foot care education elements, with shared low implementation observed in four of the six participating countries. In addition, low implementation for wearing properly sized shoes (DFCES-N11) and footwear characteristics (DFCES-N12) was observed in three countries. Consistent with these findings, studies suggest that footwear-related education may be insufficiently emphasized in routine nursing practice. A hospital-based survey of

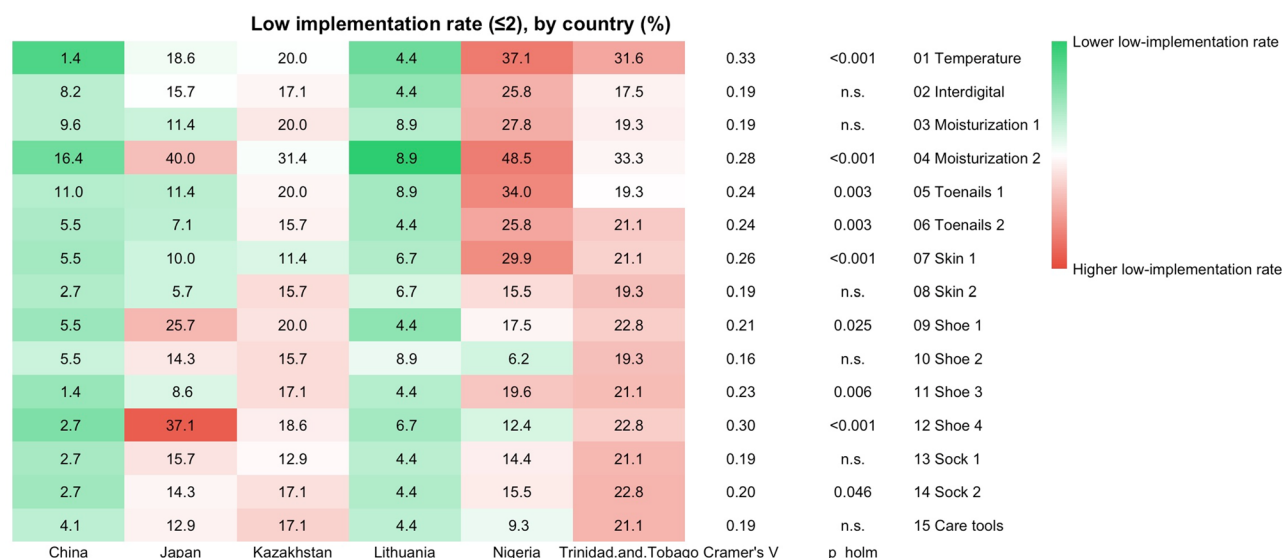


Fig. 2 Heat map of low implementation rates for DFCEs-N items among nurses in six countries ($n=412$). Note: Low implementation was defined as item scores of 1–2 (“never/rarely”). The heat map shows the percentage of nurses in each country reporting low implementation for each DFCEs-N item, compared with the pooled six-country benchmark. Cells are colored green when the low implementation rate is below the pooled six-country benchmark for that item; fewer nurses reported “never” or “rarely.” Cells are colored white when the rate is approximately equal to the benchmark. Cells are colored red when the rate is above the benchmark; more nurses reported “never” or “rarely.” Item wording is provided in Table A1. Effect-size benchmarks: 0.10/0.30/0.50 $\approx$ small/medium/large

435 nurses reported that only 11.7% provided education on footwear selection for persons with diabetes [21]. In addition, Mohamed et al. reported that 42% of nurses were not aware of the need for proper footwear [22], indicating a potential awareness gap that may hinder education delivery. Collectively, these findings align with our cross-national results and reinforce footwear guidance as a recurrently under-addressed domain of nurse-led education.

Assessment of neuropathy and vascular status is essential for identifying high-risk feet and tailoring care guidance; at the same time, foot self-care education remains relevant for all persons with diabetes and becomes particularly protective when loss of protective sensation and/or poor peripheral circulation is present [23]. Prior research [24–26] and international guidance [5] emphasize footwear- and sock-related protection as a core component of DFU prevention, including avoiding ill-fitting socks and footwear, and foreign objects inside shoes. The risks are particularly relevant for persons with peripheral neuropathy, who may not perceive early injury [23, 27]. Despite this preventive importance, the present study findings suggest that several footwear- and sock-related topics were reported with low implementation in routine education, indicating a potential misalignment between guideline priorities and nurses’ self-reported education delivery. By assessing education delivery item-by-item across six countries at the implementation level, this study extends previous work that has predominantly focused on patient risk factors [28], ulcer outcomes [29],

or aggregate elements of nurse-led education [30]. Compared with prior nursing studies that have documented variability in diabetic foot care knowledge and practice [31], the current findings add cross-national, content-specific evidence pinpointing footwear- and sock-related elements as recurring gaps.

Some factors may help interpret the recurrent under-addressing of footwear- and sock-related education elements observed in this study. First, the item-level analytic approach may have helped highlight foot education elements that are less consistently addressed in routine care, particularly those requiring individualized counseling [32], such as evaluating sock fit and footwear-related friction or pressure. These aspects may be less readily incorporated into routine education compared with more visible or procedural components of foot care. In addition, footwear- and sock-related education often involves practical skills and contextual judgment that may be inconsistently emphasized in general nursing education or continuing professional training [21, 33]. Taken together, these findings suggest a need to strengthen nurses’ delivery of detailed footwear- and sock-related education.

Cut toenails

In Kazakhstan, Nigeria, and Trinidad and Tobago, straight-cut toenail education (DFCEs-N05) and toenail inspection (DFCEs-N06) showed higher than benchmark rates of low implementation, indicating that toenail care education may represent a recurrent gap across

participating countries. Prior work from Japan has documented suboptimal nail care practices among nurses and care workers and identified nail cutting as a difficult aspect of foot care [31]. The ADA Standards of Care (2026) recommend preventive foot self-care education, including nail care, as part of routine diabetes management support [23]. Clinically, appropriate toenail trimming and inspection are important because improper cutting, particularly rounding or cutting into the nail corners, can contribute to sharp nail spicules, ingrown nails, or periungual inflammation, which may progress to skin breakdown [34]. These complications are especially concerning in diabetes with peripheral neuropathy, where early tissue damage may go unnoticed, increasing ulceration risk. Limited time and heavy workloads may reduce opportunities to deliver detailed, technique-focused education (e.g., identifying risky trimming habits and recognizing periungual problems) during routine encounters [13]. Therefore, toenail care may warrant more explicit instruction in routine nurse-led education, including correct trimming technique and regular inspection for nail changes.

Interdigital moisturization

Low implementation of advising persons with diabetes not to apply lotion between the toes (DFCES-N04) was observed in three of the six countries (Japan, Nigeria, and Trinidad and Tobago), suggesting that this interdigital “no-lotion” message may be less consistently emphasized in routine education across settings. In a survey assessing nurses’ diabetic foot management knowledge, 26% agreed that moisturizing cream should be applied to spaces between toes [21], indicating that a nontrivial proportion of nurses may hold an incorrect belief about interdigital skin care and thus may be less likely to deliver (or may unintentionally contradict) the “avoid interdigital lotion” education.

Moisturization is a key component of diabetic foot care because dry skin can lead to fissures that may serve as entry points for pathogens [6]; however, international guidance consistently distinguishes plantar/dorsal moisturization from interdigital care, explicitly advising not to apply lotions between the toes to avoid moisture retention, maceration, and infection risk [5, 35]. This distinction is clinically relevant because even minor skin infections (e.g., tinea pedis) can have disproportionate consequences in persons with diabetes, where barrier function and immune responses are often compromised [36]. Supporting this concern, an unmatched case–control study reported that using lotion between the toes was associated with higher odds of DFU (AOR = 3.35, 95% CI = 1.35–8.32) [37], underscoring that interdigital lotion use may reflect unsafe foot care practices that warrant clear preventive messaging. These findings suggest

that strengthening nurse-led education should focus not only on “use moisturizer” but also on where not to apply it, using clear phrasing.

Water temperature check

This study identified specific challenges in Nigeria and Trinidad and Tobago, where nurses reported low implementation of advising persons with diabetes to check bathwater temperature (DFCES-N01). This education is clinically important because diabetes-related peripheral neuropathy can impair thermal sensation and increase the risk of unrecognized scald injury during routine hygiene. International guidance on diabetic foot self-care includes checking bath/shower water temperature (below 37 °C) to prevent thermal injury, particularly for individuals with loss of protective sensation [5]. Although direct cross-national evidence on nurses’ delivery of education on water temperature is limited, convergent empirical findings suggest that this message may be inconsistently emphasized and incompletely adopted. A survey of healthcare professionals reported a lack of interprofessional consensus regarding bathwater temperature testing recommendations [38], implying that such education may not be uniformly prioritized across care teams. It should also be considered that Nigeria and Trinidad and Tobago have warm climates year-round (mean annual temperature ~ 27 °C) [39, 40], which may reduce the perceived need for heated water and partly contribute to lower prioritization of water temperature checks. From the patient perspective, a facility-based survey found that approximately 45.8% of persons with diabetes did not check water temperature before washing [41], suggesting that this safety message may not be consistently translated into routine self-care behaviors. Chantelau [42] reported elevated thermal detection thresholds and pain perception thresholds in neuropathic feet, suggesting increased vulnerability during exposure to heat. Practically, nurse-led education could consider providing explicit, feasible instructions, such as using lukewarm water and testing water with the elbow/forearm to reduce preventable burn risk.

In the same two countries, low implementation was also observed for advising patients to check between the toes in humid conditions (DFCES-N07), a message directly relevant to interdigital moisture retention risk. Additionally, humid tropical conditions may increase interdigital moisture and fungal maceration risk, making interdigital inspection and thorough drying especially important. This aligns with international guidance, which recommends drying the feet carefully, especially between the toes, as part of routine preventive foot self-care education [5]. Clinically, interdigital tinea pedis commonly causes maceration and fissures, which can compromise the skin barrier and increase susceptibility to secondary

infection, an issue of particular concern in persons with diabetes. Consistent with our finding that this message may be under-delivered, patient practice studies suggest that interdigital drying is frequently omitted; a survey reported that 42.4% of adults with diabetes never dried between their toes after washing [43]. Taken together, these results indicate a potential gap between guideline-stated interdigital care messages and routine education delivery. It may be easier to omit than more general hygiene education for time-limited nurses, as “between-the-toes care” is a location-specific micro-behavior. The specific drivers of under-emphasis require further study. Therefore, strengthening nurses’ foot education implementation of interdigital inspection and drying messages needs to be emphasized.

Across six countries with diverse healthcare contexts and resource settings, this study’s findings on consistent challenges in interdigital moisture management, footwear inspection, toenail trimming, and sock fitting underscore the need for improving implementation. Such targeted improvements may help bridge the gap between international recommendations and nursing practice, ensuring evidence-based and practical patient education.

Limitations

This study has some limitations. First, the data were collected via self-report questionnaires, which may be affected by information bias, including social desirability and recall bias, and potential misclassification introduced by dichotomizing Likert responses into low versus non-low implementation categories [44, 45]. Second, nurses were recruited using non-probability sampling, which may increase selection bias and limit generalizability to the broader nursing workforce within each country [46]. Third, participants were drawn from different facilities, and cross-country contrasts may partly reflect differences in care settings and service organization, such as hospital level, availability of multidisciplinary foot services, rather than country context alone. Finally, the current study did not assess nurse-self contextual factors that may influence implementation, such as formal diabetic foot care training, specialist certification, interest in foot care, and time for foot care education, which may confound country-level differences.

Future multi-country studies are warranted to collect key contextual determinants of nurse-led foot care education, enabling multilevel analyses that separate nurse-, facility-, and country-level effects. In parallel, intervention studies are warranted to test standardized, skills-based training that targets the specific content gaps identified in this study.

Conclusion

In conclusion, this study provided the first cross-national assessment of nurses’ implementation of foot care education for persons with diabetes across six countries. The study found common international gaps and substantial cross-country variation at the item level, particularly in key elements such as footwear inspection, interdigital care, and toenail care. These findings underscore the importance of strengthening the implementation of commonly under-addressed educational elements while considering local contexts. By providing comparative evidence across diverse contexts, this study may help inform context-sensitive strategies and support DFU prevention.

Abbreviations

ADA	American Diabetes Association
CFA	Confirmatory factor analyses
CVI	Content validity index
DM	Diabetes mellitus
DFCES-N	Diabetic Foot Care Education Scale nurse version
DFUs	Diabetic foot ulcers
FWER	Family-wise error rate

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12912-026-04457-7>.

Supplementary Material 1

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Author contributions

Yu Liu: Conceptualization, Methodology, Project administration, coordination of international collaboration, formal analysis, and writing of the original draft. Yukari Takeno: Supervision, Methodology, Writing, review, and editing. Olga Riklikienė, Chidum Ezenwaka, Clementina Ukamaka Nwankwo, Shengling Zhu, Dinara Makhanbetkulova: Conceptualization, Data curation, Investigation, Writing – review & editing. Rosemary Adamma Analike, Aruzhan Dabyl, Virginija Bulikaite, Xinyu Fan, Virginia Victor, and Qingbo Zhang: investigation, writing – review, and editing. All the authors have read and approved the final version of the manuscript.

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Data availability

The data supporting the conclusions of this article are included within the manuscript and the additional files.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Nagoya University, Japan (Approval No. 2023–0439), as the relevant institutional review boards in China, Lithuania, Kazakhstan, Nigeria, and Trinidad and Tobago. Participation was voluntary, and informed consent was obtained from all respondents before data collection.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form (including individual details, images, or videos).

Competing interests

The authors declare no competing interests.

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